

Conditioning as Group Action: Exact Compositional Conditioning at FiLM Cost

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Abstract

Networks are conditioned by scaling features (FiLM) or by generating weights (hypernetworks). Both learn what each condition does. Neither learns what two conditions do together unless training data shows them together. We condition by rotating features, with rotation angles a linear function of the condition. Rotations add, so combined conditions compose automatically, and the network handles combinations it never saw. **CGA** (Conditioning as Group Action) formalizes this: the condition enters the operator linearly in an abelian Lie algebra, $T(c) = P \exp(A(Wc))P^\top$, so $T(c_1+c_2) = T(c_1)T(c_2)$ holds for any weights. A diagonal algebra yields a compositional FiLM variant; the rotation algebra with scalar position is RoPE. Under a pre-registered protocol (margins, tests, and splits committed before every run; conditioning cost within $1.2\times$ FiLM), CGA posts the lowest error on never-trained condition combinations across five transformation suites, with complete seed separation ($p=9.1\times 10^{-5}$, $N=10$) and zero-shot extrapolation to combinations longer than any trained. Swapping its linear map for an MLP raises error up to 4.3×10^4 ; the most expressive baselines generalize worst. Three registered gates returned negatives that bound the method: a fit penalty on rich scenes, a content-conditioning failure (rotations move information and cannot add it), and rollouts limited by representation consistency. The negatives specify the successor. **Complex FiLM**, magnitude for content and phase for composition, beats FiLM by 36% on unseen combinations and matches it on content ($0.97\times$) under two further pre-registered gates. The same structure yields guidance without classifier-free extrapolation: powering the condition distorts $2\text{--}5\times$ less than CFG at every strength, with no second forward pass. One operator, both roles, FiLM cost.

1 Introduction

Conditioning, injecting a class, an action, a timestep, or a text embedding into a network’s computation, recurs in nearly every conditional model. Diffusion models modulate every block on the timestep and class via adaLN [2]; visual reasoning, speech, and RL systems modulate features on questions, speakers, and goals [1]; world models condition latent dynamics on actions. Almost all of this conditioning is carried by three mechanisms: concatenation, feature-wise affine modulation (FiLM), or a hypernetwork that emits weights from the condition.

These mechanisms sit at two extremes of an expressivity axis. FiLM’s diagonal transform is cheap and stable but cannot mix features. Hypernetworks can express any linear operator but pay quadratic cost and, as we show, memorize the training conditions: their error on unseen combinations of familiar conditions is worse than FiLM’s, despite $43\times$ the conditioning compute. The middle of this axis, structured operator families between diagonal and dense, is well explored for weight fine-tuning [6–9] and largely unexplored in FiLM’s role: per-sample, input-conditioned modulation of activations.

We ask which structure, at FiLM-level cost, buys compositional generalization: correct behavior on combinations of conditions never seen together in training. Our answer imports a mechanism from positional encoding. RoPE [4] and its generalization GRAPE [5] encode position n as a group action $G(n) = \exp(n\omega L)$; because position enters linearly in the Lie algebra, $G(n+m) = G(n)G(m)$ holds exactly. We transplant this construction from scalar positions in attention to arbitrary condition vectors in activation modulation.

Contributions. (1) **Framework.** CGA treats conditioning as a group representation of the condition space; FiLM, RoPE, and GRAPE drop out as special cases, and a two-line proposition makes condition composition exact for any weights (§3). (2) **Protocol.** A pre-registered, budget-fair evaluation: margins, statistical tests, and single-read splits committed to version control before every run, with a shared counter enforcing parameter and FLOP parity; we report every verdict, including negatives against our own predictions (§4). (3) **Boundary map.** Five confirmations where

conditions are transformations (up to five orders of magnitude; capacity inversion in every image suite) and three mechanistic negatives (fit, content, rollouts) that state exactly where operator guarantees stop (§5). **(4) Successor and guidance.** The negatives specify Complex FiLM, validated under two further gates as a FiLM replacement covering both conditioning roles, whose phase channel also yields guidance as an exact operator power: 2–5× less distortion than CFG with no unconditional pass (§5).

2 Related work

FiLM and conditional normalization. FiLM [1] and its normalized variants (conditional BN/LN, adaLN in DiT [2]) apply per-feature affine transforms generated from a condition. All are diagonal. We keep FiLM’s role and enrich only the operator.

Structured operators in PEFT. Low-rank, orthogonal, and butterfly operator families are well studied as weight fine-tuning: LoRA [6], OFT [7], BOFT [8], HRA [9], Group-and-Shuffle [10]; He et al. [11] give a unified view. These learn one fixed transform per task. Our setting differs on every axis that matters: per-sample, input-conditioned, activation-space, and evaluated for compositional generalization. Hypernetworks [3] generate weights from conditions and inherit the memorization behavior we quantify.

Position encodings as group actions. RoPE [4] rotates query-key pairs by angles linear in position; GRAPE [5] generalizes this to group representations with exact relative laws. The mechanism is theirs; our contribution is the transplant to arbitrary condition vectors in FiLM’s role, with a learned basis, and the budget-fair compositional evidence.

Compositional generation. Montero et al. [12] showed reconstruction-trained generative models fail on unseen factor combinations. Score and energy composition [23] composes concepts across separate conditional passes; we compose conditions inside the operator, in one pass. Symmetry-based representation learning [13, 14] argues latents should carry group actions; we supply the conditioning-side mechanism.

Nearest prior work, found adversarially. Feature-space rotations with exact composition exist for geometric transformations [17]; abelian exponential-map operators for disentanglement [22]; attribute operators for recognition [26]; multilinear attribute mixing for unseen combinations [27]; per-sample amplitude-and-phase activation modulation in periodic implicit networks [20]; and composition by complex phase addition in knowledge-graph embeddings [21]. Rotation-structured latent world models with latent-prediction losses originate with Quessard et al. [15], Caselles-Dupré et al. [16], and the homomorphism autoencoder [18], whose composition property is emergent rather than imposed; latent consistency objectives are standard in model-based RL [19]. Guidance has been moved into the conditioning path by distilling a CFG teacher [25], and CFG’s high-scale distortion is established [24]. We claim none of these components. We claim the combination: a group-structured operator in FiLM’s role at budget parity, composition imposed exactly by a linear-in-algebra map, magnitude split from phase as content and composition channels, guidance as an exact power, and verdicts from pre-registered gates.

3 The CGA framework

Let $x \in \mathbb{R}^d$ be features and $c \in \mathbb{R}^k$ a condition. FiLM computes $y = \gamma(c) \odot x + \beta(c)$; hypernetworks emit $W(c)$. All are instances of

$$y = T(c)x + \beta(c), \quad (1)$$

and they differ only in where the per-sample operator $T(c)$ comes from.

Definition 1 (Compositional conditioner). $T : \mathbb{R}^k \rightarrow \text{GL}(d)$ is compositional if it is a homomorphism from $(\mathbb{R}^k, +)$: $T(c_1 + c_2) = T(c_1)T(c_2)$ and $T(0) = I$.

Proposition 1 (Exact composition by construction). Let $\mathfrak{a}$ be an abelian subalgebra of $\mathfrak{gl}(d)$, $A : \mathbb{R}^m \rightarrow \mathfrak{a}$ linear, $W \in \mathbb{R}^{m \times k}$, and P orthogonal. Then $T(c) = P \exp(A(Wc)) P^\top$ is a compositional conditioner for every W, P . If $\mathfrak{a}$ is skew-symmetric, $T(c)$ is orthogonal.

Proof. $\exp(X)\exp(Y) = \exp(X+Y)$ for commuting X, Y ; $A \circ W$ is linear; conjugation preserves products; $\exp(0) = I$. $\square$

Composition is usually a behavior the network must learn from data. Here it is a property of the parametrization. It holds at initialization, during training, and far outside the training distribution. Training only decides which directions the condition rotates and how fast.

Special cases. Choices of $(\mathfrak{a}, c, P)$ recover deployed mechanisms:

$\mathfrak{a}$	condition	P	recovered mechanism
$\text{diag}(\mathbb{R}^d)$	c	I	<i>exponential FiLM</i> : $y = e^{Wc} \odot x$ (compositional)
$\mathfrak{so}(2)^{d/2}$	$c = n$ (scalar position)	I	RoPE [4]
$\mathfrak{so}(2)^{d/2}$	$c = n$, learned planes	learned	multiplicative GRAPE [5]
$\mathfrak{so}(2)^{d/2}$	general $c \in \mathbb{R}^k$	structured	this work (FiLM’s role)
$\text{diag} \times \mathfrak{so}(2)^{d/2}$	general c	canonical	Complex FiLM (§5)

Standard FiLM reaches the same diagonal operators through an MLP head $\gamma(c)$. The operators match; the composition law is missing, and §5 shows the missing law is exactly what fails on unseen combinations.

Cost. With $\mathfrak{a} = \mathfrak{so}(2)^{d/2}$, $\exp$ is closed-form planar rotation: $3d$ FLOPs, no matrix exponential. The basis P is the whole budget question. A dense learned P costs $4d^2$ FLOPs per sample, which is $1.52\times$ FiLM’s entire conditioning path at $d=128$. Two fixes work. A structured orthogonal P (five block-diagonal Cayley layers with fixed shuffles, following Group-and-Shuffle) brings the total to $1.11\times$ FiLM, with 4,800 parameters against the 8,128 dimensions of $\text{SO}(128)$; that is enough, because P only needs the planes the condition acts on. Inside a transformer, no explicit P is needed: the adjacent dense projections absorb the basis, which is how RoPE deploys. The operator equals the identity at initialization and at $c=0$, and every singular value is 1. Unit tests pin all three properties. Complex FiLM extends the algebra to $\text{diag} \times \mathfrak{so}(2)^{d/2}$: each feature pair is multiplied by $m e^{i\theta}$, with magnitude from FiLM’s expressive head (the content channel) and phase linear in the condition (the composition channel), at $0.84\times$ FiLM’s cost in our harnesses.

4 Pre-registered, budget-fair protocol

Design. Every suite is pre-registered: success margins, statistical tests ($N=10$ seeds, one-sided Mann–Whitney $p \leq 0.01$, Cliff’s $|\delta| \geq 0.474$), and data splits were fixed and committed to version control before each run. Each test split is read once, after all selection on a validation split. All arms share one backbone and one training recipe. A shared counter enforces the budget: our conditioning path stays within $1.20\times$ FiLM’s FLOPs and $1.05\times$ the smallest hypernetwork baseline’s parameters; baselines may be larger, and reach $48\times$ our parameter count. Deviations are dated amendments in the committed specifications, plus one accounting erratum that our own audit caught and that downgraded a favorable verdict.

Threats to validity, and the design answer. The benchmarks are self-constructed, so we built them to remove judgment from scoring: dSprites and 3D Shapes are deterministic factor-to-image maps, so every held-out condition has a unique ground-truth image and pixel error is an objective compositional metric, with no perceptual score and no human rating. Pre-registration removes outcome-contingent analysis; the verdicts below include three negatives, one of which falsified our own headline prediction. Every number in this paper regenerates from committed logs with one command per suite, so replication at our scale costs one training run.

Arms. FiLM; conditional LayerNorm; Concat-MLP; a dense hypernetwork $I+\Delta W(c)$; a dynamic low-rank map $I+A(c)B(c)^\top$; the Lie operator; Complex FiLM (both variants); and an MLP-head ablation of our own method (same basis, same operator family, more compute, no linearity).

Table 1: Error on never-trained condition combinations: mean (seed sd) over 10 seeds. CGA is lowest in every column, with complete seed separation against the strongest hypernetwork baseline ($p=9.1\times 10^{-5}$, $\delta=-1.0$). S1–S3b pass the full pre-registered gate; S4 fails a separate fit criterion (text). S7 is the Complex FiLM rerun of the S3 harness.

Conditioner	S1	S2	S3	S3b	S4	S7
FiLM	$1.3 (0.1)\times 10^{-2}$	$1.6 (0.04)\times 10^{-2}$	$3.0 (0.2)\times 10^{-3}$	$5.9 (0.2)\times 10^{-3}$	$4.5 (1.0)\times 10^{-3}$	$3.1 (0.2)\times 10^{-3}$
Concat-MLP	$1.5 (0.3)\times 10^{-2}$	$1.1 (0.4)\times 10^{-2}$	$2.9 (0.1)\times 10^{-3}$	$6.3 (0.2)\times 10^{-3}$	$5.4 (5.2)\times 10^{-3}$	–
Cond-LN	$2.4 (0.1)\times 10^{-2}$	$2.7 (0.05)\times 10^{-2}$	$3.2 (0.2)\times 10^{-3}$	$6.1 (0.3)\times 10^{-3}$	$8.6 (13.1)\times 10^{-3}$	–
Hypernetwork	$1.9 (0.3)\times 10^{-3}$	$3.4 (0.9)\times 10^{-3}$	$3.8 (0.3)\times 10^{-3}$	$7.0 (0.4)\times 10^{-3}$	$9.3 (2.3)\times 10^{-3}$	$3.8 (0.2)\times 10^{-3}$
Dyn. low-rank	$5.7 (0.5)\times 10^{-3}$	$6.1 (0.4)\times 10^{-3}$	$5.1 (0.9)\times 10^{-3}$	$8.1 (0.4)\times 10^{-3}$	$2.0 (0.4)\times 10^{-2}$	–
MLP-head (abl.)	–	$1.1 (0.2)\times 10^{-3}$	$2.7 (0.2)\times 10^{-3}$	$5.0 (0.2)\times 10^{-3}$	$3.3 (0.8)\times 10^{-3}$	–
Complex FiLM (lin.)	–	–	–	–	–	$2.0 (0.1)\times 10^{-3}$
Complex FiLM	–	–	–	–	–	$2.0 (0.2)\times 10^{-3}$
Lie (ours)	$7.4 (2.0)\times 10^{-4}$	$3.3 (2.2)\times 10^{-8}$	$1.8 (0.1)\times 10^{-3}$	$4.0 (0.1)\times 10^{-3}$	$1.2 (0.2)\times 10^{-3}$	$1.8 (0.1)\times 10^{-3}$

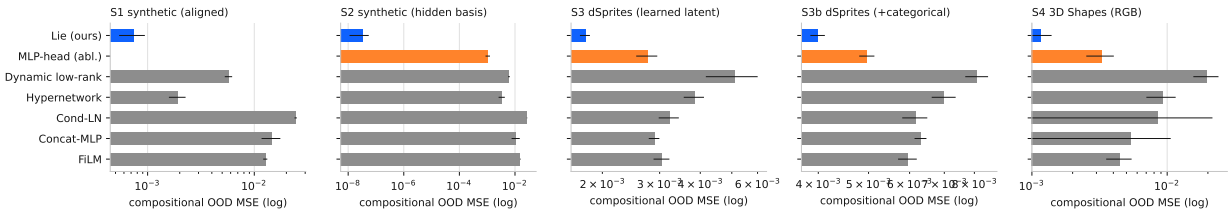


Figure 1: Error on unseen condition combinations by conditioner (log scale; blue = ours, orange = MLP-head ablation), across the five transformation suites.

Suites. **S1/S2 (synthetic):** learn $y = M(c)x$, where c selects a combination of 8 rotation primitives; S2 hides the primitives behind a random basis and tests all 56 never-trained *triples*. **S3/S3b (dSprites) and S4 (3D Shapes):** image transformation: encode a real image, apply $T(\Delta)$ on the learned latent for a factor change Δ , decode, and score pixel error against the true transformed image; OOD means never-trained *types* of two-factor changes; S3b/S4 add a categorical shape factor. **S5 (content):** a small diffusion transformer generates dSprites images from full factor specifications. **S6 (rollouts):** a world model applies the conditioning arm recurrently, one action per step; horizons 10 and 20 after training on 3. **S7 (successor):** Complex FiLM through the S3 and S5 harnesses. **S8 (guidance):** condition-dropout training, then CFG at strength w for FiLM against condition powering for Complex FiLM, scored against deterministic ground truth per strength.

5 Results

(1) When conditions form a group, composition is solved. On S2, CGA reaches 3.3×10^{-8} error on unseen pairs and 5.3×10^{-8} on never-trained triples; the best baseline sits five orders of magnitude higher and every baseline degrades as combinations lengthen. Learned angles match the true generators to 2×10^{-4} rad, so the operator is directly readable (Figure 3).

(2) The advantage survives learned representations. On dSprites every arm fits the training distribution equally well, isolating composition. CGA’s error grows $1.25\times$ from training to unseen combinations ($1.63\times$ with the categorical factor); every baseline grows $2.1\text{--}3.3\times$. CGA beats the strongest baseline by 53.8% (42.9%) at $39\times$ fewer conditioning FLOPs.

(3) The linear map is the mechanism. The ablation keeps the basis and the operator family, spends more compute, and swaps only the linear map for an MLP. It loses by $4.3\times 10^4\times$ on synthetic triples and $1.55\times$ on dSprites.

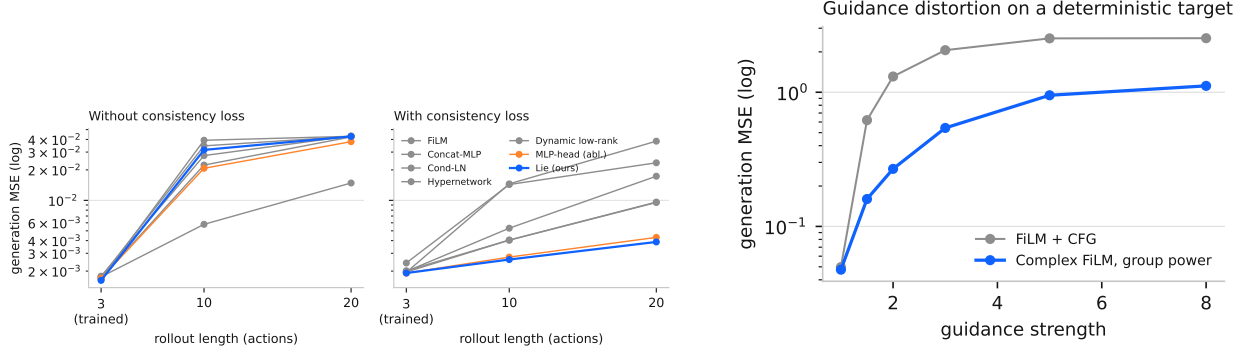


Figure 2: **Left (S6/S6’)**: rollout error versus horizon, without and with the latent-consistency loss; with it, the isometric operator stays flat where every baseline drifts. **Right (S8)**: generation error versus guidance strength; CFG’s linear extrapolation distorts $50.7\times$ from strength 1 to 8, condition powering $23.7\times$, staying $2\text{--}5\times$ lower throughout.

(4) More capacity makes it worse. The two hypernetwork arms can express any operator. They post the worst generalization gaps on images ($2.7\text{--}26\times$); their capacity memorizes the training combinations.

(5) Three registered negatives. On 3D Shapes, CGA has the best OOD numbers (87.5% margin) but fits training $1.46\times$ worse than the best memorizer, tripping our no-regression gate. In the content suite, rotation conditioning fails (0.31 vs. film’s 0.04) and the ablation fails identically: a rotation moves information and cannot add any, so it cannot carry “generate this image.” In the rollout suite, every arm compounds error at the same rate; each step adds a small mismatch between predicted and true latent, which a contraction shrinks and an isometry carries forever.

(6) A consistency loss completes the rollout recipe. Training with $\lambda\|T(a)z_s - z_{s+1}\|^2$, applied to every arm equally, inverts the picture: the rotation operator’s horizon-20 error goes flat at 0.0039 (growth $2.0\times$) against the hypernetwork’s 0.038 (growth $19.8\times$) and FiLM’s 0.017 (Figure 2, left). The loss is standard in model-based RL [19], and rotation transitions with latent-prediction objectives appear in the homomorphism autoencoder [18]; our contribution is the controlled decomposition at matched budget, plus a null that sharpens it: fixed uniform contraction at any rate from 0.003 to 0.1 leaves horizon-20 error unchanged, so the correction must be adaptive. That suite’s registered gate scores KILL on a separate single-step criterion, and we report it so.

(7) The negatives specify the fix, and it works. Complex FiLM multiplies feature pairs by $m e^{i\theta}$. Under two pre-registered gates it beats FiLM by 36% on unseen change types at fit parity and $0.84\times$ FiLM’s cost, and matches FiLM on content generation at $0.97\times$ its error (pre-registered non-inferiority margin $1.10\times$), where the pure rotation failed by $7\times$. A linear magnitude channel fails (0.19): content needs the expressive head. FiLM is the real part of the working operator; the whole complex number does both jobs.

(8) Guidance without the second pass. CFG’s distortion at high scales is established [24]; our deterministic instrument isolates it against an exact alternative. CFG’s error grows $50.7\times$ from strength 1 to 8; powering the condition through the phase channel grows $23.7\times$ ($p=1.6\times 10^{-4}$), stays $2\text{--}5\times$ lower at every strength above 1, matches at native strength ($0.96\times$), and needs no unconditional pass and no distillation [25] (Figure 2, right). Neither method improves over strength 1 here: the target is deterministic, so there is no distribution to sharpen, exactly as the suite’s registered caveat stated.

6 Practical guidance

Three rules for practitioners, from the eight suites. **Use phase conditioning when your conditions are changes that combine.** Edit deltas, attribute offsets, pose or camera adjustments, applied once. Cost matches FiLM, fit matches, and behavior on never-trained combinations is 43% to $10^5\times$ better; FiLM’s failure there is silent, and the guarantee removes

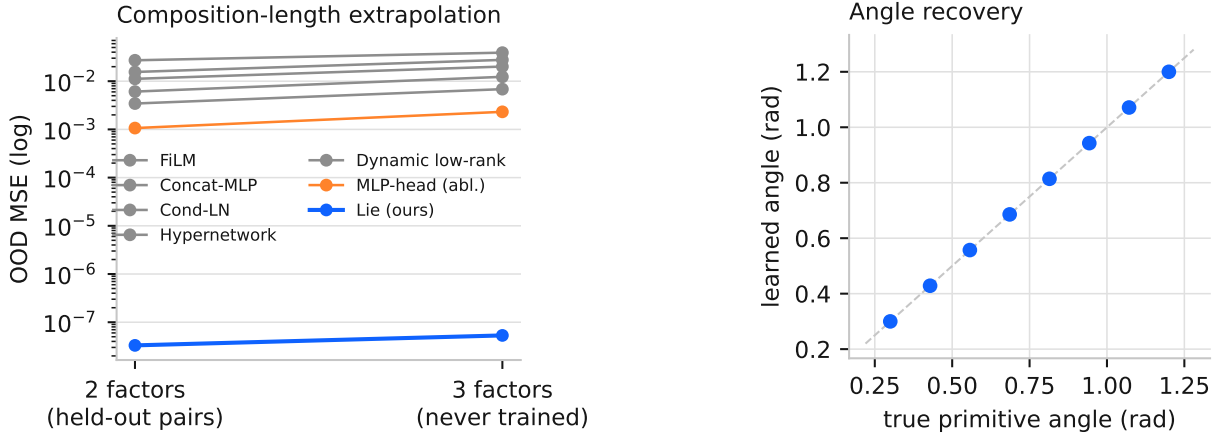


Figure 3: **Left (S2):** extrapolation to composition length 3, never trained; ours is flat at 10^{-8} and every baseline degrades. **Right (S1):** learned singleton angles recover the true generators to 2×10^{-4} rad.

that failure class. **For conditions that say what to generate, use Complex FiLM or keep FiLM.** Pure rotation fails there ($7\times$); Complex FiLM matches FiLM on content ($0.97\times$) while keeping the composition wins, so it is the candidate drop-in at our evidence scale. **In rollouts, pair an isometric operator with a consistency loss.** Without the loss, all conditioners compound error at the same rate; with it, the rotation operator stays flat where FiLM grows $4.5\times$ and a hypernetwork $10\times$.

7 Limitations

All evidence comes from controlled 64×64 benchmarks with known factors, width-128 models, and a single conditioning site; nothing here is validated at production scale, and the natural deployment experiment is the adaLN site of a text-conditioned diffusion transformer. Learned latents do not support exact group action: the real-image advantage is a reliable $2\times$, not the synthetic $10^5\times$. S4 shows orthogonality can price in-distribution fit on rich scenes. The guidance instrument isolates distortion and cannot measure the distribution-sharpening benefits CFG provides on multimodal targets. Conditions far from any group action, such as open-ended text, remain untested; the learned-condition-geometry extension is future work.

Reproducibility statement. Every suite is pre-registered in committed specifications; every number in this paper regenerates from append-only logs with one command per suite; unit tests pin the operator invariants (identity at initialization and at $c = 0$, orthogonality, exact composition) and the budget counters; all amendments and one accounting erratum are disclosed in the repository, which we release with the paper.

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